PathForge

An AI Career Roadmap for ECE Students — built with Lovable

Built with: Lovable (No-Code AI)

Team

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Problem Statement

Engineering students often struggle with career confusion due to lack of structured learning path.

- **No clear direction**

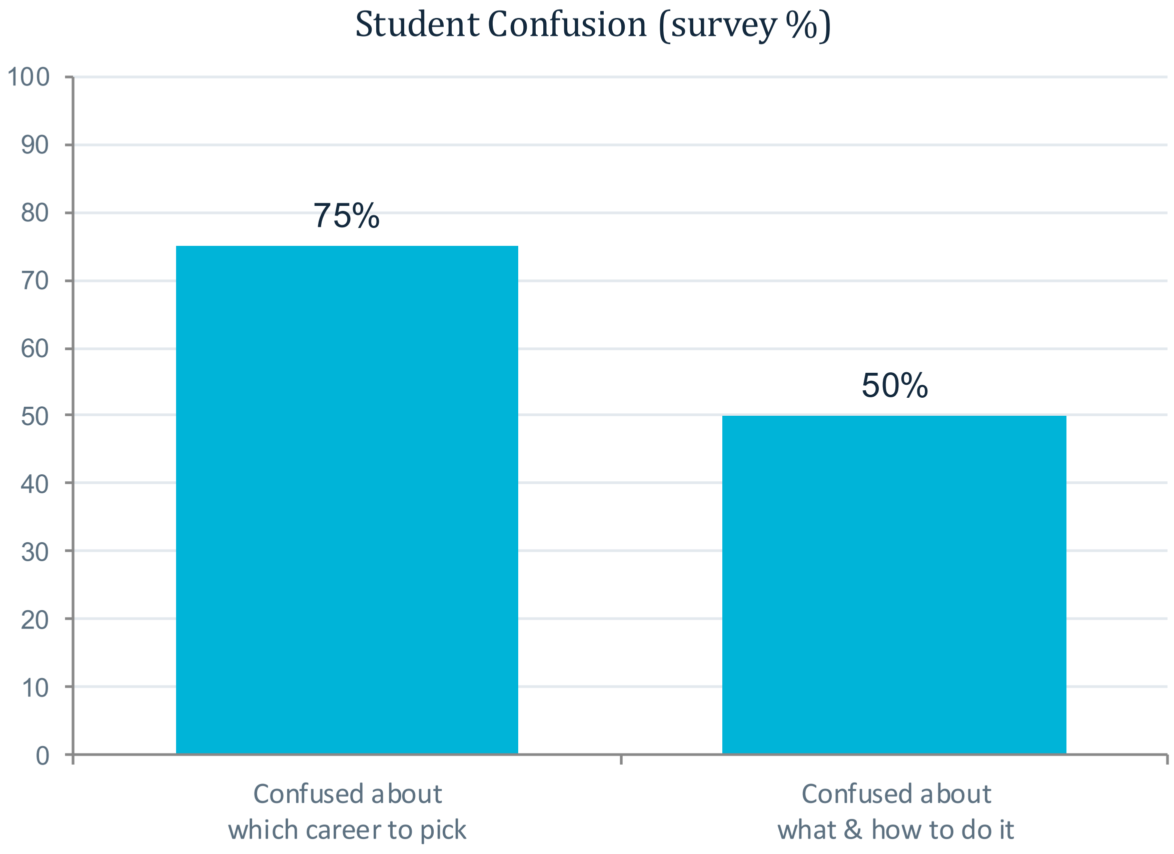
  ECE students face 10+ career paths (VLSI, embedded, IoT, RF) with no easy way to compare fit.

- **Information is everywhere , direction is rare**

  We don’t lack information , Its dumping on our head.

- **Lack of clarity , direction**

  Students has no clarity, direction to reach our goal.



Source: internal peer survey among ECE undergraduates

Our Approach

1

Speed

Lovable turns plain-English prompts into a working full-stack web app in minutes, not weeks.

2

No backend coding

Database, routing and UI are generated together — the team focused on logic and content, not syntax.

3

Fast iteration

Every change (add a feature, tweak a flow) is just a new prompt, then an instant live preview.

4

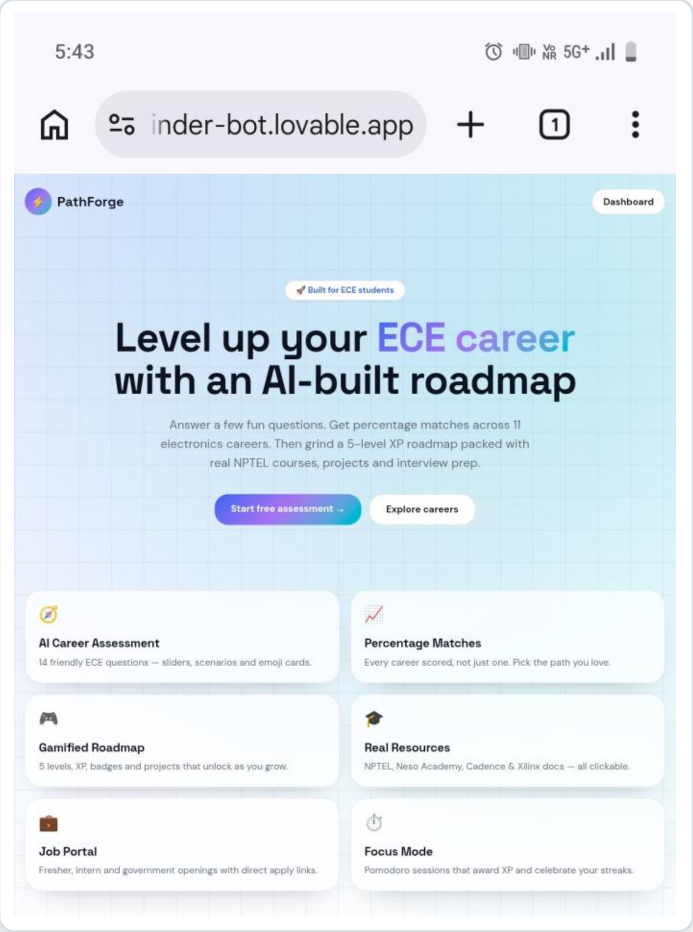
Team-friendly

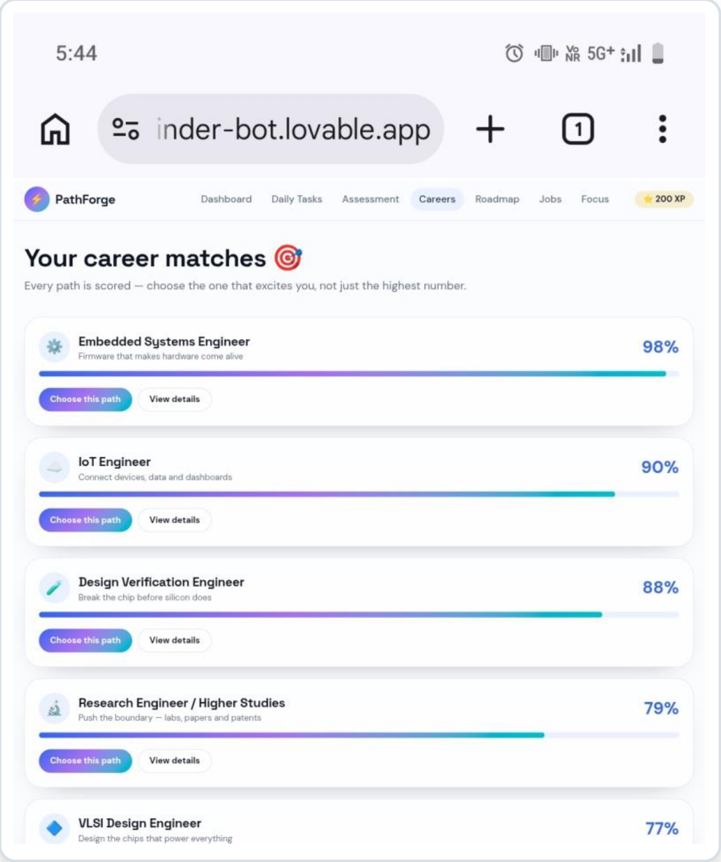
Non-programmers on the team could contribute directly to building, not just planning.

Working

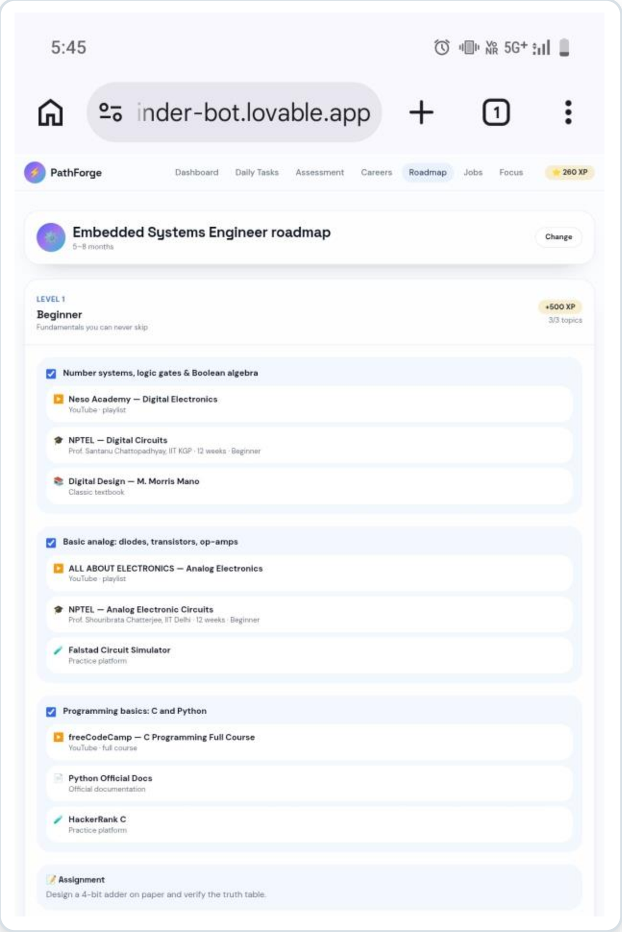
Prompt → Generate → Test → Deploy

1. Prompt in Lovable
2. Generate app
3. Test each flow
4. Live webpage





Careers — tested output after training the flow



Home — generated from a single prompt

Roadmap — deployed on ece-pathfinder-bot.lovable.app

Challenges & Solutions

What went wrong, and how we fixed it

Career-match accuracy	Challenge: Early prompts gave vague, one-size-fits-all results.	Fix: Rewrote the assessment prompt with 14 specific ECE questions and weighted scoring across 11 careers.
Content overload	Challenge: Dumping every resource link made pages cluttered.	Fix: Curated a fixed shortlist per topic (NPTEL, Neso Academy) instead of open-ended lists.
Keeping users engaged	Challenge: A static roadmap felt like a to-do list, easy to abandon.	Fix: Added XP, levels, streaks and badges so progress feels visible and rewarding.

Tools & Technologies

What we built with,

Lovable

Core no-code AI builder — generated the full app from prompts

AI prompting

Iterative prompt refinement to shape features and flows

Canva

Supporting visuals and graphics for slides and mockups

PowerPoint

This presentation and demo materials

Lovable Hosting

Live deployment at ece-pathfinder-bot.lovable.app

Mobile browser

Cross-device testing of the deployed webpage

Key Learnings

What this project taught us

- **No-code lowers the barrier, not the ambition**

  We built a full multi-page app with gamification and no traditional coding — no-code doesn't mean simple.

- **Data quality decides output quality**

  Vague prompts and thin content gave weak results; specific, well-structured input made the AI's output usable.

- **Teamwork across skill levels**

  Members with different comfort levels in tech could all contribute — writing content, testing flows, designing prompts.

- **Iteration beats a perfect first try**

  The best version came from repeatedly testing, breaking, and re-prompting — not from getting it right the first time.

Closing Statement

Thank you for the opportunity to build and present PathForge.

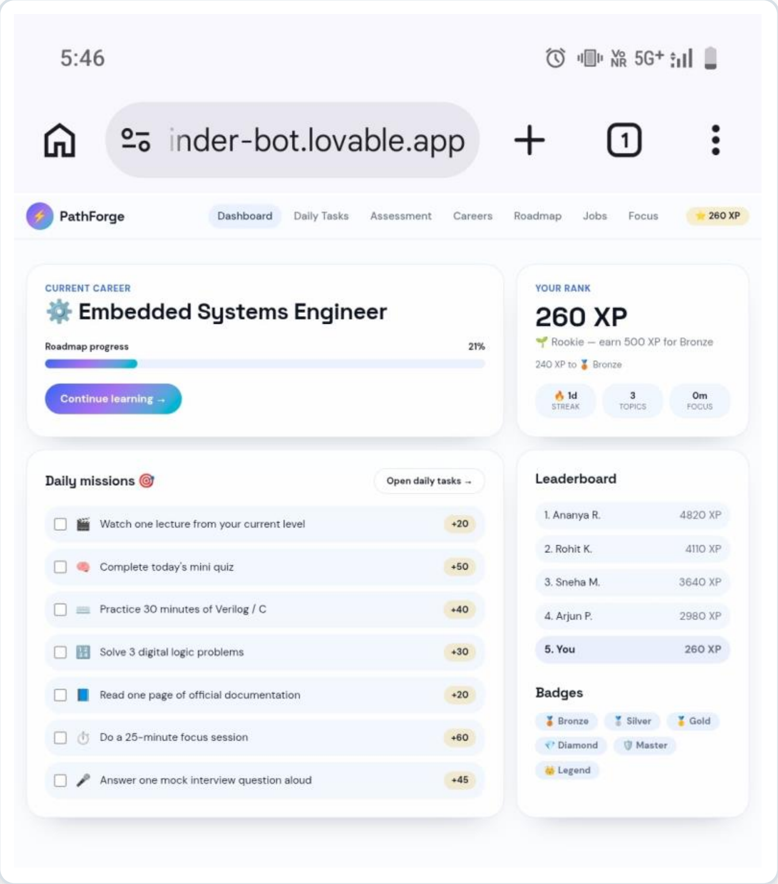
Future scope

- Expand beyond ECE to other engineering branches
- Add mentor connect and peer discussion features
- Scale the job portal with real-time listings

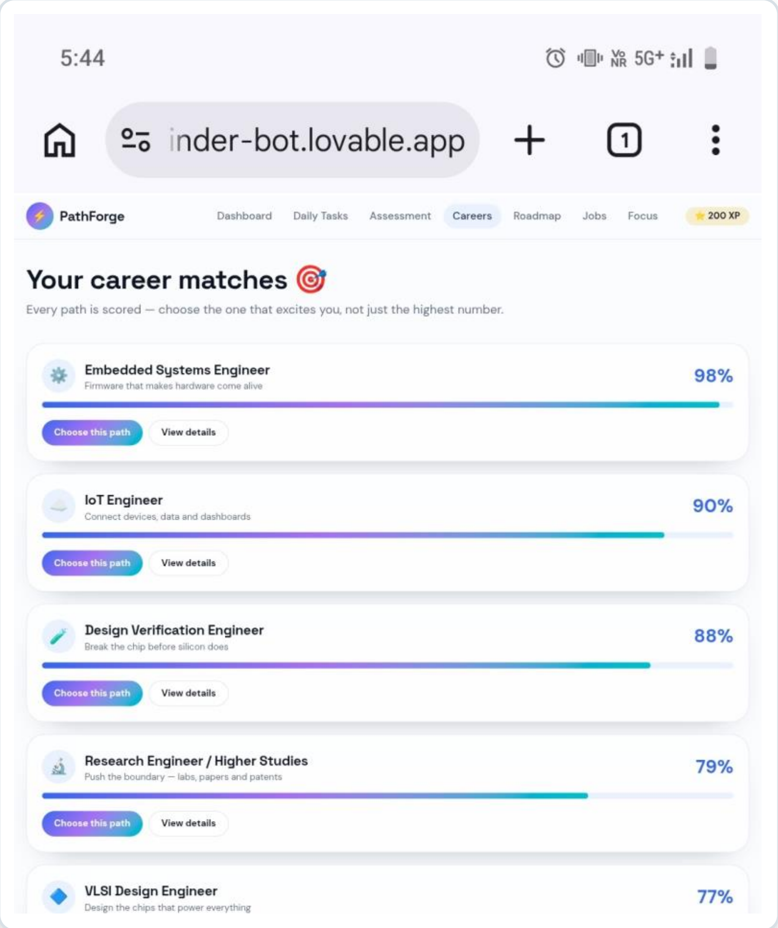
Built as a team effort — proof that a good idea and a no-code tool can go a long way.

Live Demo

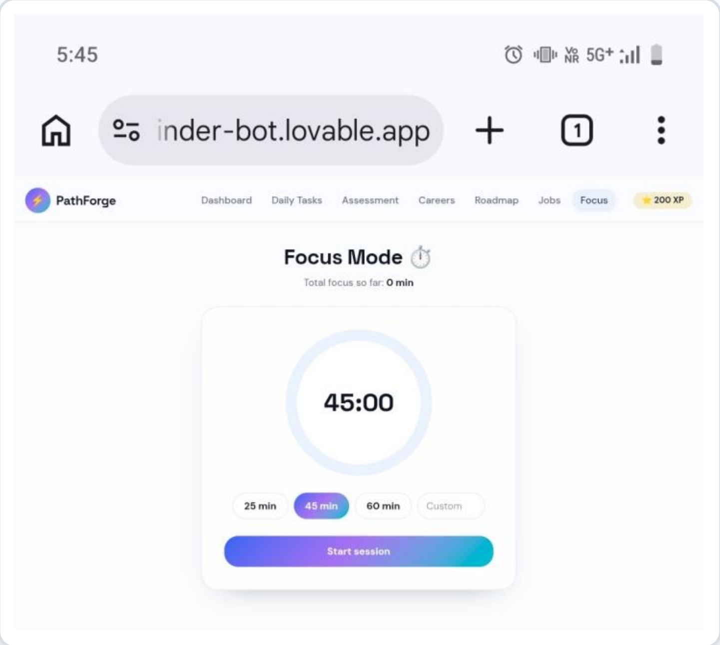
https://ece-pathfinder-bot.lovable.app/



Input: student profile & progress



Output: scored career matches



Behind the scenes: Focus Mode timer

Questions & Answers

Ask us about: why Lovable, how we handled challenges, and what we'd improve next.

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